1 An aeroplane accelerates along a horizontal runway before take-off.
The aeroplane accelerates for 35 s. The speed of the aeroplane when it takes off is 72 m / s.

Fig. 1.1 shows how the speed of the aeroplane varies between time t = 0 and t = 35 s.

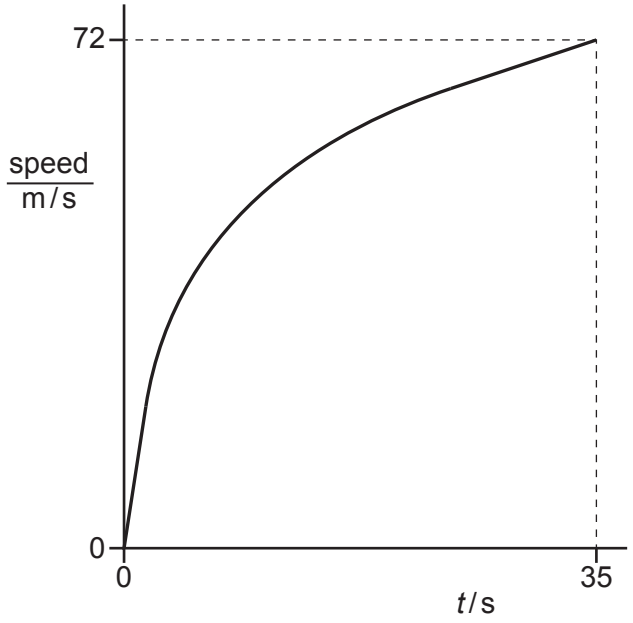


Fig. 1.1

(a) Define acceleration.

.....

..... [1]

(b) **(i)** Calculate the average acceleration of the aeroplane between t = 0 and t = 35 s.

acceleration = ... [1]

(ii) The combined mass of the aeroplane, its passengers and its fuel on take-off is 1.1×10^5 kg.

Calculate the average resultant force on the aeroplane between t = 0 and t = 35 s.

force = ... [2]

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(iii) The force provided by the engines of the aeroplane is constant.

Give **one** possible explanation for the change in acceleration of the aeroplane between $t = 0$ and $t = 35$ s.

.....

..... [1]

(iv) On Fig. 1.2, sketch a graph to show how the acceleration of the aircraft varies between $t = 0$ and $t = 35$ s.

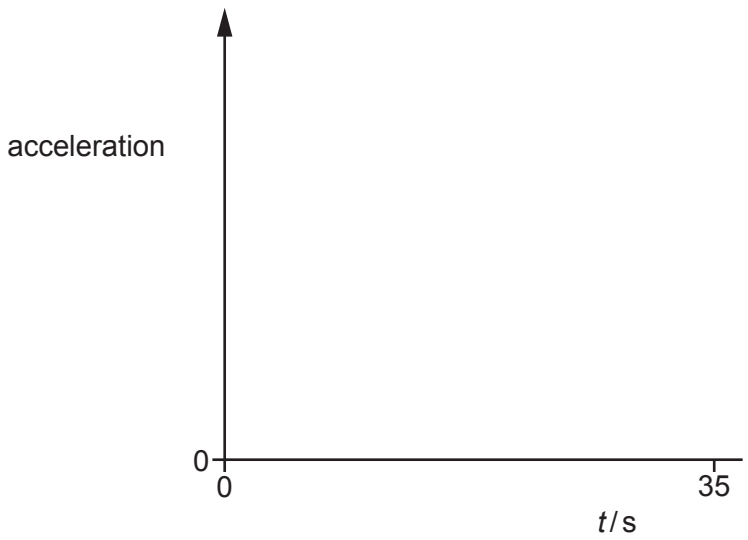


Fig. 1.2

[3]

[Total: 8]